



Certificate of Compliance

Certificate: 1072937 (111996)

Master Contract: 188782

Project: 2739771

Date Issued: June 25, 2014

Issued to: Cavagna Group S.P.A.
5 Via Matteotti
Viadana di Calvisano, Brescia 25012
Italy
Attention: Luigi Tomaselli

The products listed below are eligible to bear the CSA Mark shown with adjacent indicators 'C' and 'US' for Canada and US or with adjacent indicator 'US' for US only or without either indicator for Canada only.



James Sekerak

Issued by: James Sekerak

PRODUCTS

CLASS 3331 82 - REGULATORS (GAS) - Gas Appliance Pressure-Certified to U.S. Standards

CLASS 3331-02 **REGULATORS (GAS) – Gas Appliance Pressure**

Model Number or Type

For Use With Natural Gas

Pressure Regulators – 1/2 psi

96(0,1)(L)(F), Inlet Size: 3/8, Outlet Sizes: 3/8, 1/2"-20 UNF

97(0,1)(L)(F), Inlet Size: 3/8, Outlet Size: 3/8

97(0,1)(L)(F), Inlet Size: 1/2, Outlet Size: 1/2

98(0,1)(L)(F), Inlet Size: 1/2, Outlet Size: 1/2

98(0,1)(L)(F), Inlet Size: 3/4, Outlet Size: 3/4



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Pressure Regulators – 2 psi

96 2(L)(F), Inlet Size: 1/4, Outlet Size:1/4

96 2(L)(F), Inlet Size: 3/8, Outlet Size:3/8

For Use With Propane Gas or Liquefied Petroleum Gases

Pressure Regulators – 1/2 psi

96(5,6)(L)(F), Inlet Size: 1/4, Outlet Size:1/4

96(5,6)(L)(F), Inlet Size: 3/8, Outlet Size:3/8

97(5,6)(L)(F), Inlet Size: 3/8, Outlet Size:3/8

97(5,6)(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98(5,6)(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98(5,6)(L)(F), Inlet Size: 3/4, Outlet Size:3/4

For Use With Natural Gas and

For Use With Propane Gas or Liquefied Petroleum Gases

Pressure Regulators – 1/2 psi

96 9(L)(F), Inlet Size: 1/4, Outlet Size:1/4

96 9(L)(F), Inlet Size: 3/8, Outlet Size:3/8

97 9(L)(F), Inlet Size: 3/8, Outlet Size:3/8

97 9(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98 9(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98 9(L)(F), Inlet Size: 3/4, Outlet Size:3/4



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Trade Name: "O"

CLASS 3331-82 REGULATORS (GAS) – Gas Appliance Pressure
Model Number or Type

For Use With Natural Gas

Pressure Regulators – 1/2 psi

96(0,1)(L)(F), Inlet Size: 3/8, Outlet Sizes:3/8, 1/2"-20 UNF

97(0,1)(L)(F), Inlet Size: 3/8, Outlet Size:3/8

97(0,1)(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98(0,1)(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98(0,1)(L)(F),Inlet Size: 3/4, Outlet Size:3/4

Pressure Regulators – 2 psi

96 2(L)(F), Inlet Size: 1/4, Outlet Size:1/4

96 2(L)(F), Inlet Size: 3/8, Outlet Size:3/8

For Use With Propane Gas or Liquefied Petroleum Gases

Pressure Regulators – 1/2 psi

96(5,6)(L)(F), Inlet Size: 1/4, Outlet Size:1/4

96(5,6)(L)(F), Inlet Size: 3/8, Outlet Size:3/8

97(5,6)(L)(F), Inlet Size: 3/8, Outlet Size:3/8



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97(5,6)(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98(5,6)(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98(5,6)(L)(F), Inlet Size: 3/4, Outlet Size:3/4

For Use With Natural Gas and

For Use With Propane Gas or Liquefied Petroleum Gases

Pressure Regulators – 1/2 psi

96 9(L)(F), Inlet Size: 1/4, Outlet Size:1/4

96 9(L)(F), Inlet Size: 3/8, Outlet Size:3/8

97 9(L)(F), Inlet Size: 3/8, Outlet Size:3/8

97 9(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98 9(L)(F), Inlet Size: 1/2, Outlet Size:1/2

98 9(L)(F), Inlet Size: 3/4, Outlet Size:3/4

Trade Name: “O”

APPLICABLE REQUIREMENTS

ANSI Z21.18-2007 •CSA 6.3-2007, ANSI Z21.18a-2010 •CSA 6.3a-2010, ANSI Z21.18b-2012 •CSA 6.3b-2012 - Pressure Regulators for Gas Appliances



Descriptive Report and Test Results

MASTER CONTRACT: 188782

REPORT: 1072937

PROJECT: 2739771

- Edition 1:** 1999-10-06; Report 111996-01 – Cleveland
Issued by James McAnlis; Reviewed by Trevor W. Perera
- Edition 2:** 1999-02-24; Project 111996-04 – Cleveland
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- Edition 3:** 1999-03-17; Project 111996-05 – Cleveland
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- Edition 5:** 1999-08-06; Project 111996-14 – Cleveland
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- Edition 6:** 2000-02-01; Project 1053862 – Cleveland
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- Edition 8:** 2000-03-28; Project 1072937 – Cleveland
Issued by Andris Udris; Reviewed by Trevor W. Perera
- Edition 9:** 2000-03-28; Project 1072937 – Cleveland
Issued by Andris Udris; Reviewed by Trevor W. Perera
- Edition 10:** 2000-10-18; Project 1136254 – Cleveland
Issued by Andris Udris; Reviewed by Trevor W. Perera
- Edition 11:** 2001-02-20; Project 1176859 - Cleveland
Issued by Valdis Udris
- Edition 12:** 2001-04-09; Project 1193562 – Cleveland
Issued by Richard Weiser
- Edition 13:** 2002-11-20; Project 1380341 – Cleveland
Issued by Valdis Udris
- Edition 14:** January 6, 2005; Project 2257492 – Cleveland
Issued by James Sekerak
Addition of Figures 108 to 110
- Edition 15:** June 25, 2014; Project 2739771 – Cleveland
Issued by James Sekerak
Addition of Figures 111 to 175

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Supplement to Certificate of Compliance – Page 1
Description and Tests – Pages 1 to 12
Att1 Manufacturing and Production Test Plan – 1 to 30
Att2 Drawings – 1 to 175

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PRODUCTS

CLASS 3331-02 REGULATORS (GAS) – Gas Appliance Pressure

<u>Model Number or Type</u>	<u>Inlet Size</u>	<u>Outlet Size</u>
For Use With Natural Gas		
Pressure Regulators – 1/2 psi		
96(0,1)(L)(F)	3/8	3/8
	3/8	1/2"-20
97(0,1)(L)(F)	3/8	3/8
	1/2	1/2
98(0,1)(L)(F)	1/2	1/2
	3/4	3/4
Pressure Regulators – 2 psi		
96 2(L)(F)	1/4	1/4
	3/8	3/8
For Use With Propane Gas or Liquefied Petroleum Gases		
Pressure Regulators – 1/2 psi		
96(5,6)(L)(F)	1/4	1/4
	3/8	3/8
97(5,6)(L)(F)	3/8	3/8
	1/2	1/2
98(5,6)(L)(F)	1/2	1/2
	3/4	3/4
For Use With Natural Gas and For Use With Propane Gas or Liquefied Petroleum Gases		
Pressure Regulators – 1/2 psi		
96 9(L)(F)	1/4	1/4
	3/8	3/8
97 9(L)(F)	3/8	3/8
	1/2	1/2
98 9(L)(F)	1/2	1/2
	3/4	3/4
Trade Name: "O"		

CLASS 3331-82 REGULATORS (GAS) – Gas Appliance Pressure

<u>Model Number or Type</u>	<u>Inlet Size</u>	<u>Outlet Size</u>
For Use With Natural Gas		
Pressure Regulators – 1/2 psi		
96(0,1)(L)(F)	3/8	3/8
	3/8	1/2"-20
97(0,1)(L)(F)	3/8	3/8
	1/2	1/2

98(0,1)(L)(F)	1/2	1/2
	3/4	3/4
Pressure Regulators – 2 psi		
96 2(L)(F)	1/4	1/4
	3/8	3/8

For Use With Propane Gas or Liquefied Petroleum Gases

Pressure Regulators – 1/2 psi		
96(5,6)(L)(F)	1/4	1/4
	3/8	3/8
97(5,6)(L)(F)	3/8	3/8
	1/2	1/2
98(5,6)(L)(F)	1/2	1/2
	3/4	3/4

**For Use With Natural Gas and
For Use With Propane Gas or Liquefied Petroleum Gases**

Pressure Regulators – 1/2 psi		
96 9(L)(F)	1/4	1/4
	3/8	3/8
97 9(L)(F)	3/8	3/8
	1/2	1/2
98 9(L)(F)	1/2	1/2
	3/4	3/4

Trade Name: “O”

APPLICABLE REQUIREMENTS

ANSI Z21.18-2007 •CSA 6.3-2007, ANSI Z21.18a-2010 •CSA 6.3a-2010, ANSI Z21.18b-2012 •CSA 6.3b-2012
- Pressure Regulators for Gas Appliances

MARKINGS

The manufacturer is required to apply the following markings:

- Products shall be marked with the markings specified by the particular product standard.
- Products certified for Canada shall have all Caution and Warning markings in both English and French.

Additional bilingual markings not covered by the product standard(s) may be required by the Authorities Having Jurisdiction. It is the responsibility of the manufacturer to provide and apply these additional markings, where applicable, in accordance with the requirements of those authorities.

All markings shall be permanent cast, mechanically stamped or using affixed label.

- Manufacturer's name, trademark or symbol
- Outlet pressure adjustment range
- Rated inlet pressure
- CSA C/US monogram
- Direction of flow
- Regulator class
- Date code
- Model Number

ALTERATIONS

Material (NBR EN 549 60 SH.) of diaphragm and washer may be supplied by Tecnogomma International Srl.

Project 2257492 – Plug Cover material is aluminum

FACTORY TESTS

The submitter shall use a program to qualify raw materials, parts, assemblies and purchased components.

The submitter shall ensure that the following factory tests are conducted at the frequency specified and the results are documented and made available for review by CSA field services representatives:

<u>Test - ANSI Z21.18</u>	<u>Frequency</u>
Leakage (2.4) - at room temperature Pass/Fail Criteria The regulator shall not leak externally at a rate in excess of 200 cubic centimeters per hour at pressure.	100%
Outlet pressure setting or corresponding flow rate Pass/Fail Criteria The pressure regulator, if non-adjustable, shall have an outlet pressure that is within ± 0.3 inches water column or ± 5 percent of the manufacturer's specified outlet pressure.	100%
Continued operation (2.13) Pass/Fail Criteria The pressure regulator shall withstand 100,000 cycles of opening and closing of the regulator valve, without any mechanical failure, impairment of operation, apparent damage, and without the development of leakage. At the completion of 100,000 cycles, the regulator shall comply with the room temperature tests specified in Leakage, and P4-1 shall be within $\pm 75\%$ of P3-1, P4-2 shall be within $\pm 70\%$ of P3-2, P4-3 shall be within $\pm 60\%$ of P3-3, P4-4 shall be within $\pm 50\%$ of P3-4 and T4 shall not be greater than T3 + 1 minute. After 1 second of gas flow the outlet pressure shall not exceed 120 percent of P4-SS. After 4 seconds of gas flow the outlet pressure shall be at least 0.2 inch water column.	Annually
Outlet pressure adjustment range (2.8) Pass/Fail Criteria The pressure regulator, if adjustable, when set to the minimum setting, shall have an outlet pressure that is equal to or less than the manufacturer's specified minimum outlet pressure. The pressure regulator, if adjustable, when set to the maximum setting, shall have an outlet pressure that is equal to or greater than the manufacturer's specified maximum outlet pressure. The pressure regulator, if non-adjustable, shall have an outlet pressure that is within ± 0.3 inches water column or ± 5 percent of the manufacturer's specified outlet pressure.	Annually
Range of regulation capacity (2.9) Pass/Fail Criteria The pressure regulator, if non-adjustable, shall have an outlet pressure that does not deviate more than ± 20 percent, at any flow rate within the manufacturer's specified range of regulation, when subjected to an inlet pressure range that is 1.0 inches water column above the specified outlet pressure up to 21 inches water column.	Annually

Integrity of operation (2.12)

Annually

Pass/Fail Criteria

- P1a-1 shall be within $\pm 75\%$ of P1-1, P1a-2 shall be within $\pm 70\%$ of P1-2, P1a-3 shall be within $\pm 60\%$ of P1-3, P1a-4 shall be within $\pm 50\%$ of P1-4 and T1a shall not be greater than T1 + 1 minute. After 1 second of gas flow the outlet pressure shall not exceed 120 percent of P1a-SS. After 4 seconds of gas flow the outlet pressure shall be at least 0.2 inch water column.
- P2a-1 shall be within $\pm 75\%$ of P2-1, P2a-2 shall be within $\pm 70\%$ of P2-2, P2a-3 shall be within $\pm 60\%$ of P2-3, P2a-4 shall be within $\pm 50\%$ of P2-4 and T2a shall not be greater than T2 + 1 minute. After 1 second of gas flow the outlet pressure shall not exceed 120 percent of P2a-SS. After 4 seconds of gas flow the outlet pressure shall be at least 0.2 inch water column.
- P1b-1 shall be within $\pm 75\%$ of P1-1, P1b-2 shall be within $\pm 70\%$ of P1-2, P1b-3 shall be within $\pm 60\%$ of P1-3, P1b-4 shall be within $\pm 50\%$ of P1-4 and T1b shall not be greater than T1 + 1 minute. After 1 second of gas flow the outlet pressure shall not exceed 120 percent of P1b-SS. After 4 seconds of gas flow the outlet pressure shall be at least 0.2 inch water column.
- P2b-1 shall be within $\pm 75\%$ of P2-1, P2b-2 shall be within $\pm 70\%$ of P2-2, P2b-3 shall be within $\pm 60\%$ of P2-3, P2b-4 shall be within $\pm 50\%$ of P2-4 and T2b shall not be greater than T2 + 1 minute. After 1 second of gas flow the outlet pressure shall not exceed 120 percent of P2b-SS. After 4 seconds of gas flow the outlet pressure shall be at least 0.2 inch water column.

Strength (2.5.1 and 2.5.2)

Annually

Pass/Fail Criteria

- The regulator shall be capable of withstanding, without deformation, breakage or leakage, a turning effort of 220 (1/4") inch-pounds, 280 (3/8") inch-pounds, 375 (1/2") inch-pounds and 560 (3/4") inch-pounds. The regulator shall then comply with the leakage test.
- The regulator shall be capable of withstanding a bending moment of 16 (1/4"), 18 (3/8") 20 (1/2") and 22 (3/4") pounds without leakage in excess of 200 cubic centimeters per hour at 21.0 inches water column pressure applied at the inlet.

Leakage (2.4)

Annually

Pass/Fail Criteria

- The pressure regulator shall not leak externally at a rate in excess of 200 cubic centimeters per hour at 21.0 inches water column inlet pressure, at the manufacturer's specified minimum and maximum operating temperatures.

NOTE: It is the submitter's responsibility to ensure the factory tests are conducted as specified above and the complete test records are to be available for review by CSA field service representatives. Test methods should relate back to those in the performance standard.

SPECIAL INSTRUCTIONS FOR FIELD SERVICES

No special instructions for field services are required.

COMPONENT SPECIAL PICKUP

No component special pickup is required.

DESCRIPTION

MODEL NUMBER BREAKDOWN

<u>Character(s)</u>	<u>Designation</u>
96,97,98	Type/Series Code
0	Outlet Pressure - in. w.c.: 2.8 to 5.2 (96,97) (Nat.)
	" " 3.3 to 6.0 (98) "
1	" " 4.0 to 8.0 (96-98) "
2	" " 7.0 (96) "
5	" " 6.0 to 10.0 (96-98) (LP)
6	" " 8.0 to 12.0 (96,97) "
	" " 5.0 to 12.0 (98) "
9	" " 4.0/10.0 (96) (Conv.)
	" " 4.0,5.0/10.0 (97) "
	" " 5.0/10.0 (98) "
0*	OARA
G*	Sandblasted
15 & 5*	Inlet & Outlet Connections - in. NPT(F): 1/4 x 1/4 (96)
41 & 6*	3/8 x 3/8 (96,97)
33 & 9*	1/2 x 1/2 (97,98)
38 & H*	3/4 x 3/4 (98)
None	Vent Size (1.4 mm)
L	Limiter (0.35 mm)
F	Fixed Cap

* for Part/Product No. designation only

SPECIFICATIONS

Mounting Position:	Multi-poise
Ambient Operating Temperature Range:	-40°F to 205°F (-40°C to 96°C)
Capacities* (Btu/hr)	
Pressure Drop:	48,000 (96 series) 100,000 (97 series) 230,000 (98 series)
Range of Regulation:	150 to 65,000 (96 series) 150 to 120,000 (97 series) 150 to 250,000 (98 series)
Max. Regulation M & P Load:	50,000 (96 series) 90,000 (97 series) 240,000 (98 series)
Max. Individual Load:	30,000 (96 series) 40,000 (97 series) 40,000 (98 series)
Max. Variable Load:	130,000 (97 series)

* Based on a gas having a heating value of 1,000 Btu per cu. ft. and a specific gravity of 0.64 at a pressure drop of 1.0 in. w.c.

ACCESSORY TYPE

Pressure Regulator for Gas Appliances

DRAWING INDEX

PART/DRAWING NO.	REV.	PAGE NO.	TITLE
M&P Plan		1 to 30	Manufacturing and Production Test Plan
Type 98(0,1,5,6,9) (L)			
980 000 03	1	001	Whole General Assy 3/4"
" " 04	1	002	" " " 1/2"
989 " 01	1	003	" " " 3/4" Convertible
" " 02	1	004	" " " 1/2" Convertible
980 020 01		005	Body 3/4"
" " 02		006	Cover OARA
" " 03		007	Body 1/2"
976 030 01		008	Spring 3.3"-6"
981 " 01		009	" 4"-8"
985 " 01		010	" 6"-10"
986 " 01		011	" 5"-12"
989 " 01		012	" LPG Convertible
970 040 03		013	O-Ring Closing Nut
" 050 01		014	Closing Nut
" 090 01		015	Adjusting Screw
" " 05		015-1	" "
" " 03		016	Spring Pressing LPG Convertible
" " 04		017	Plug of the Vent Hole
980 040 01		018	Diaphragm
" " 02		019	Rubber Washer
" 050 02		020	Piston
" 060 01		021	Washer
" 090 01		022	Diaphragm Plate
980 100(01,02)	2	023,024	Label 980(L)
981 " (01,02)	1	025,026	" 981(L)
985 " (01,02)	1	027,028	" 985(L)
986 " (01,02)	1	029,030	" 986(L)
989 " (01,02)	1	031,032	" 989(L)
Type 96(0,1,2,5,6,9) (L)			
960 000 03	1	033	Whole General Assy 3/8"
" " 04	1	034	" " " 1/4"
" " 05	1	096	" " " 3/8" Fixed
969 " 01	1	035	" " " 3/8" Convertible
" " 02	1	036	" " " 1/4" Convertible
960 020 01		037	Body 3/8"
" " 02		038	Cover OARA
" " 03		039	Body 1/4"
960 030 01		040	Spring 2.8"-5.2"
961 " 01		041	" 4"-8"
965 " 01		042	" 6"-10"
966 " 01		043	" 8"-12"
969 " 01		044	" LPG Convertible
960 040 01		045	Diaphragm
" " 02		046	Rubber Washer
" " 03		047	O-Ring Closing Nut
" 050 01		048	Closing Nut
" " 02		049	Piston
" 060 01		050	Washer
" 090 01		051	Adjusting Screw

" " 06		051-1	" "
" " 02		052	Diaphragm Plate
" " 03		053	Spring Pressing LPG Convertible
" " 04		054	Plug of the Vent Hole
962 100 03		095	Disc 962 Fixed
960 100(01,02)	1	055,056	Label 960(L)
961 " (01,02)	1	057,058	" 961(L)
962 " (01,02)	1	059,060	" 962(L)
965 " (01,02)	1	061,062	" 965(L)
966 " (01,02)	1	063,064	" 966(L)
969 " (01,02)	1	065,066	" 969(L)

Type 97(0,1,5,6,9) (L)

970 000 03	1	067	Whole General Assy 1/2"
" " 04	1	068	" " " 3/8"
979 " 01	1	069	" " " 1/2" Convertible
" " 02	1	070	" " " 3/8" Convertible
970 020 01		071	Body 1/2"
" " 02		072	Cover OARA
" " 03		073	Body 3/8"
970 030 01		074	Spring 2.8"-5.2"
971 " 01		075	" 4"-8"
975 " 01		076	" 6"-10"
976 " 02		077	" 8"-12"
979 " 01		078	" LPG Convertible
970 040 01		079	Diaphragm
" " 02		080	Rubber Washer
" " 03		081	O-Ring Closing Nut
" 050 02		082	Piston
" 060 01		083	Washer
" 090 02		084	Diaphragm Plate
970 100(01,02)	1	085,086	Label 970(L)
971 " (01,02)	1	087,088	" 971(L)
975 " (01,02)	1	089,090	" 975(L)
976 " (01,02)	1	091,092	" 976(L)
979 " (01,02)	1	093,094	" 979(L)
IS 0005	3	097	Instructions - Safety Warning (English)
IS 0005	3	098	" " " (French)
96009007		099	Cover Plug For Stab. 960
96009008		100	Cover Plug For Stab. 960
97009006		101	Cover Plug For 97 and 98 Series
97009007		102	Cover Plug For 97 and 98 Series w/CSA Mark

Type 966L – Alt. Silicone Diaphragm

44-4-390-0016	0	103	966L Assembly
44-4-360-0014	0	104	" "
04-4-110-0037	0	105	Silicone Washer
09-4-110-0022	0	106	Silicone Diaphragm
34-4-310-0022	9	107	Label
19-1-950-0052	0	108	S/I Closing Nut – Type 97
44-1-590-0007	3	109	Stab. Type 98 (Ref. 9800G38HL) w/ Safety
19-4-110-0043	0	110	Closing Nut

Type 96

PART/DRAWING NO.	REV.	PAGE NO.	TITLE
00-1-150-0055	0	111	Body for "OARA" Regulator 1/4" NPT
00-1-150-0141	0	112	Body for "OARA" Regulator 1/4" NPT (w1/8")
00-4-121-0019	2	113	Body for "OARA" Regulator 3/8" NPT
00-4-121-0033	1	114	Body for "OARA" Regulator G1/4" NPT
00-4-121-0034	1	115	Body for "OARA" Regulator G1/4" NPT x 3/8"
00-4-121-0045	1	116	Body for "OARA" Regulator 3/8" NPT /1/2-20
00-4-150-0027	0	117	Body for "OARA" Regulator 3/8" NPT Machined
00-4-150-0028	0	118	Body for "OARA" Regulator 3/8" NPT Machined
00-4-150-0063	0	119	Body for "OARA" Regulator 3/8" NPT Machined
00-4-150-0111	1	120	Body for "OARA" Regulator 1/4" x 3/8" NPT
01-1-150-0183	0	121	Cover for "OARA" Reg. Type 96
01-4-121-0009	1	122	Cover for "OARA" Reg. Type 96 Casting
01-4-150-0009	0	123	Cover for "OARA" Reg. Type 96 M/C Hole Ø0.35
01-4-150-0010	0	124	Cover for "OARA" Reg. Type 96 M/C Hole Ø1.4
03-4-110-0003	0	125	Piston
04-4-110-0033	0	126	Rubber Washer
04-4-110-0037	1	127	Silicone Washer
04-4-110-0501	0	128	O-ring Ø6.4 x 1.9
04-4-110-0502	0	129	O-ring Ø1.78 x 12.42
07-1-110-0273	2	130	Spring Stabilizer 96 (Convertible LPG)
07-1-110-0293	1	131	Spring Stabilizer 96B
07-1-110-0304	1	132	Spring Stabilizer 966
07-4-110-0032	1	133	Spring Stabilizer 961
07-4-110-0033	1	134	Spring Stabilizer 965
07-4-110-0088	2	135	Spring Stabilizer 96B
08-1-110-0051	0	136	Spring Retainer GPL for Stabilizer Type 96
08-1-110-0052	1	137	Adjusting Screw for Stabilizer Type 96
08-4-110-0005	0	138	Closed Adjusting Screw for Stabilizer Type 96
09-1-950-0081	0	139	S/I Diaphragm (Stab. Type 96)
09-1-950-0084	0	140	S/I Diaphragm (Stab. Type 96 – Silicon)
09-4-110-0007	0	141	Diaphragm
09-4-110-0022	0	142	Silicone Diaphragm
11-4-110-0007	0	143	Diaphragm Plate
15-4-110-0002	0	144	Washer
18-1-310-0021	0	145	Label (Ink Stamp) for Stabilizer Type 96
18-1-110-1959	0	146	Disc Plug Stab Stabilizer Type 96 (White)
18-1-110-1960	0	147	Disc Plug Stab Stabilizer Type 96 (Black)
18-1-110-1961	0	148	Disc Plug Stab Stabilizer Type 96 (Yellow)
18-1-110-1962	0	149	Disc Plug Stab Stabilizer Type 96 (Green)
18-1-110-1963	0	150	Disc Plug Stab Stabilizer Type 96 (Blue)
18-1-310-0006	0	151	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0011	0	152	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0012	0	153	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0013	0	154	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0014	0	155	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0019	0	156	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0020	0	157	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0027	0	158	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0028	0	159	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0032	0	160	Label (Ink Stamp) for Stabilizer Type 96

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PART/DRAWING NO.	REV.	PAGE NO.	TITLE
18-1-310-0057	0	161	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0058	0	162	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0060	0	163	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0072	0	164	Label (Ink Stamp) for Stabilizer Type 96
18-1-310-0145	0	165	Label (Ink Stamp) for Stabilizer Type 96
19-1-110-2014	0	166	Protection Cap
19-1-950-0048	0	167	S/I Plug – 1/8 NPT
19-1-950-0050	0	168	S/I Closing Nut – Type 96
19-4-110-0031	0	169	Plug – 1/8 NPT
19-4-110-0032	0	170	Vent Hole Plug
19-4-110-0034	0	171	Closing Nut
19-4-110-0034	0	172	News Stabilizers OARA (Ref. IS0005)
44-C-490-0005	1	173	Stab. Type 97 with Safety
44-C-590-0017	1	174	Stab. Type 98 with Safety
44-C-490-0029	1	175	Stab. Type 98 with Safety

TESTS

Project No. 111996-01

Tests were conducted at CSA Cleveland Laboratory. The actual test data results are maintained in CSA Cleveland facility. Testing was conducted on model no. 98.

Satisfactory results were obtained on the following tests:

ANSI Z21.18 Part I. Construction

Part II. Performance

- 2.1 General
- 2.2 Test and Reference Gases
- 2.3 Test Pressures
- 2.4 Leakage
- 2.5 Strength and Deformation
- 2.6 Mounting Regulator for Test - DNA - No tests in this section
- 2.7 Pressure Drop Capacity
- 2.8 Outlet Pressure Range
- 2.9 Range of Regulation Capacity
- 2.10 Regulators Designated to Operate at Pilot Flow Rate
- 2.11 Regulators for Use on Domestic Gas Ranges
- 2.12 Integrity of Operation
- 2.13 Continued Operation
- 2.14 Resistance to Permanent Damage of Excessive Supply Pressure
- 2.15 Vent Limiter
- 2.16 Marking Material Adhesion and Legibility

Part III. Negative Pressure Regulators - Performance - DNA - Regulator is not a negative pressure regulator

Part IV. Manufacturing and Production Tests

Project No. 2257492

No tests were conducted during this project.

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Project No. 2739771

No tests were conducted during this project.